

MODULE 1B

Definition, Objectives & IR vs. Data Retrieval

A one-hour session on how Information Retrieval is formally defined, what it's actually trying to achieve, and how it fundamentally differs from database-style retrieval.

Reference: Manning, Raghavan & Schütze — Introduction to Information Retrieval (Cambridge University Press)

Quick Recap: The Motivation

1

Data Overload

We've moved from data scarcity to data overload — the challenge is finding what's relevant, not finding data at all.

2

Unstructured Data

Most real content — emails, articles, chats — has no fixed schema, unlike a SQL table.

3

Browsing Breaks Down

Manually browsing folders worked at small scale; it collapses once collections reach millions of documents.

4

Information Need

IR exists to satisfy what a user actually needs — not just to answer their literal typed query.

Today's Roadmap

01

Defining IR Formally

Unpacking Manning & Raghavan's textbook definition, phrase by phrase.

12 min

02

A Short History

How we got from library catalogues to modern ranked search.

8 min

03

The 5 Objectives

What every IR system is actually trying to achieve, with real examples.

20 min

04

IR vs. Data Retrieval

The core conceptual divide, a comparison table, and a worked example.

15 min

05

Discussion & Recap

A class activity, key takeaways, and space for questions.

5 min

Defining Information Retrieval

“

Information retrieval is finding material (*usually documents*) of an unstructured nature (*usually text*) that satisfies an information need from within large collections (*usually stored on computers*).

— Manning, Raghavan & Schütze, *Introduction to Information Retrieval*, Ch. 1

Notice this definition is doing three things at once:

- It names WHAT we search — material, usually documents
- It names the CHALLENGE — unstructured, usually free text
- It names the GOAL — satisfying a need, not just matching words, at scale

Breaking Down the Definition

Documents

The basic unit retrieved — a webpage, a PDF, an email, even a single paragraph. Deliberately flexible.

Example: Google indexes a full webpage; a plagiarism checker indexes a single paragraph.

Unstructured Data

No fixed schema, unlike a database row. Most real-world text — articles, chats, reports — has no predictable field structure.

Example: There's no SQL column for 'documents about AI affecting jobs.'

Information Need

The real underlying gap in someone's knowledge — richer than the few keywords they actually type.

Example: Your real need is nuanced; your typed query is a compressed, lossy version of it.

A Short History of IR

Pre-1960s

Library Catalogues

Manual card catalogues, subject
dexes — human-curated, browsable
at small scale.

1970s-80s

Vector Space Model

Salton's model: represent
documents/queries as vectors, rank by
similarity.

Today

ML-Driven Ranking

Machine learning and neural models
refine relevance far beyond keyword
overlap.

1960s-70s

Boolean Retrieval

Early computerized search using
AND/OR/NOT logic — exact matching,
no ranking.

1990s

Web Search Engines

Scale explodes; PageRank and link
analysis emerge to rank the open web.

The 5 Objectives of an IR System

Every IR system — from Google to your college's internal search — is judged against the same five goals. We'll take each one in depth over the next few slides.

1 Satisfy the real information need — not just literal keywords

2 Maximize Precision — keep irrelevant results out

3 Maximize Recall — don't miss relevant documents

4 Rank by Relevance — best results surface first

5 Scale Efficiently — handle huge, messy data in real time

Objective 1 & 2: Need and Precision

01

Satisfy the Real Need

A query is only a compressed, imperfect signal of what a user actually wants. The system's real job is to satisfy the underlying need, not just match the literal words typed.

Example: You need help for tomorrow's viva on ranking, but you only type "IR ranking explained simple." A good system still finds beginner-friendly material — not just pages containing those exact words.

02

Maximize Precision

Precision = the fraction of retrieved documents that are actually relevant. High precision means the user isn't wading through junk to find what they need.

Example: If you search and get 20 results but only 4 are useful, precision is low — most users won't scroll past the first page anyway, so precision at the top matters most.

Objective 3 & 4: Recall and Ranking

03

Maximize Recall

Recall = the fraction of ALL relevant documents in the collection that were actually retrieved. High recall means the system isn't missing important material.

Example: A legal search tool that misses even one relevant case law citation could cost a lawyer their argument — in legal or medical search, recall is often prioritized over precision.

04

Rank by Relevance

With potentially millions of matches, dumping an unordered pile on the user is useless. The system must order results so the most likely-useful ones appear first.

Example: Google's entire product is really its ranking algorithm — anyone can find pages containing your keywords; the hard part is deciding which one to show first.

Objective 5: Scale Efficiently

05

Scale Efficiently

The system must handle huge, messy, ambiguous collections — and still respond in real time. This is why indexing (covered in Module 1) exists at all — linear scanning breaks down at scale.

Example: Scanning 100TB of data on every query is impossible; an inverted index turns a search into a near-instant lookup instead.

The Core Goal

Tie all five objectives together into one sentence:

“Deliver the right information, to the right user, at the right time — with minimum effort on the user's part.”

Every algorithm you study for the rest of this course — indexing, TF-IDF, ranking models — is really just a strategy for getting closer to this one sentence.

Why These Objectives Matter

53%

of e-commerce shoppers abandon a search
after poor results

Precision failure = lost sales

1 in 3

clinical searches miss a relevant study without
good recall

Recall failure = missed evidence

<1s

is the expected response time for most
modern search systems

Scale failure = users simply leave

Illustrative figures used to anchor the concept — not exact citations. The point to remember: each objective failing has a real, measurable cost.

IR vs. Data Retrieval: The Core Divide

Data Retrieval

Built for structured data with a known schema — think of a students table with fixed columns.

```
WHERE cgpa > 9
```

Either a row matches, or it doesn't. No ambiguity, no ranking needed.

Information Retrieval

Built for unstructured, free-form text where there's no fixed field to query against.

```
"articles about AI and jobs"
```

No exact field exists for this. The system must estimate relevance and rank its best guesses.

IR vs. Data Retrieval: The Comparison

Feature	Data Retrieval (DB)	Information Retrieval (IR)
Matching	Exact match (SQL)	Partial / approximate match
Model	Deterministic	Probabilistic
Results	Unordered set	Ranked list by relevance
Inference	Deductive	Inductive

Memory hook for the exam:

A database GUARANTEES a correct exact-match answer with no order. IR GUESSES at relevance and always ranks its best guess first. That single sentence covers all four rows above.

Worked Example: Same Need, Two Systems

Scenario: A student wants to find "articles about how artificial intelligence affects entry-level jobs."

Database Approach

```
SELECT * FROM articles  
WHERE topic = 'AI jobs';
```

Fails immediately — no such exact column or value exists. Even if it did, it would miss every article that discusses this topic using different wording.

IR Approach

```
query: "AI effect on jobs"
```

Tokenizes and expands the query, matches partial/synonymous terms across the whole collection, then ranks candidates by estimated relevance — returning a useful ordered list.

Quick Class Activity

Turn to the person next to you and decide together:

"Finding an exact duplicate file on a hard disk using its hash value" — is this Data Retrieval or Information Retrieval? Why?

Answer & Reasoning

Data Retrieval. There's an exact match (the hash either matches or it doesn't), no ranking is needed, and no relevance judgment is involved — it's a deterministic yes/no lookup, exactly like a SQL WHERE clause.

Summary: Key Takeaways

1

IR is formally defined as finding unstructured material that satisfies an information need, from large collections.

2

Every IR system is judged against 5 objectives: satisfying the real need, precision, recall, ranking, and scaling efficiently.

3

The core goal in one line: right information, right user, right time, minimum effort.

4

IR differs from Data Retrieval on 4 fronts: matching, model, results, and inference — DB guarantees exact answers, IR ranks best guesses.

Questions?

Thank you for your attention.

Next: The Retrieval Process, Indexing, and the Query Processor — Module 2.